ISABELLA BAUER

AUTHENTICATION

Individual blog project

AUTHENTICATION

WHAT

Proving who you are

WHY

Only the right people can do certain things

TODAY

How sign-in works, where login data is stored, and how to keep it secure

THREE PLACES DATA CAN BE

Browser

What the visitor sees.



Server

code that runs hidden from the user



Database

where data is saved permanently

MAKING AN ACCOUNT

1.
Browser sends
Invite code, username, name and password

→

2.
Server checks
Sign up rules

→

3.
Server hashes
Password is scrambeled.

→

4.
Database saves
A new row in the users table.

Hash = a one-way scramble of the password.
Invite code = the host code makes you a host, the guest code a guest

SIGNING IN AND STAYING SIGNED IN

1.
Compare
Server hashes what you typed and compares it with the saved hash

2.
Session
A match saves a session row with a random id

3.
Cookie
The browser gets the id in a cookie

4.
Every click
Cookie → server finds the session → knows who you are

Session = a row in the database that says “this person is signed in”
Cookie = a small piece of data the browser saves and sends back

WHAT I USED

Browser

What the visitor sees.



Server

code that runs hidden from the user



Database

where data is saved permanently

Server: Vercel

Database: Neon

THE INVITE CODE

```
// The invite code decides the role
let role;
if (inviteCode === HOST_INVITE_CODE) role = 'host';
else if (inviteCode === GUEST_INVITE_CODE) role = 'guest';
else return fail(400, { ...values, error: 'That invite code is not valid.' });

if (await getUserByUsername(username)) {
    return fail(400, { ...values, error: 'That username is already taken.' });
}
```

AN ACCOUNT = ONE ROW IN USERS

id integer	username text	display_name text	password_hash text	role text	created_at timestamp w...
1	richard	Richard	cannot-sig…	host	2026-11-01 15:46:22.91…
2	eva	Eva	cannot-sig…	host	2026-11-01 15:46:22.95…
3	sofie	Sofie	cannot-sig…	guest	2026-11-01 15:46:22.99…
4	jonas	Jonas	cannot-sig…	guest	2026-11-01 15:46:23.03…
5	hanna	Hanna	cannot-sig…	guest	2026-11-01 15:46:23.07…
8	isabellabauer	Isabella	jq73Aho.ZL…	guest	2026-11-01 16:14:17.13…
19	eduardo	Eduardo	AipGEvQdz0…	guest	2026-11-01 18:20:34.18…
20	isabella	isabella	tAJO2pP4rp…	host	2026-11-01 18:58:56.40…
21	erik	erik	$2b$12$.TSEwQTco4R57hY…	guest	2026-11-02 10:23:29.31…

id = a number from the database
username = unique, no two the same
dispaly_name = the name shown on posts
password_hash = only the hash, never the password
role = host or guest
created_at = when the account was made

SIGNING IN AND STAYING SIGNED IN

zsh .env.example

.env.example

```
# Copy this file to `.env` and fill in real values.
# `.env` is ignored by git, this example file is not, so only use placeholders here.

# Neon PostgreSQL connection strings (Neon dashboard -> Connect).
# DATABASE_URL is the pooled one (host name ends in -pooler), used by the website.
DATABASE_URL="postgresql://USER:PASSWORD@HOST-pooler/DATABASE?sslmode=require"
# Optional: the direct connection, used by `npm run seed` for creating tables.
# If it's missing, the seed script uses DATABASE_URL instead.
DATABASE_URL_UNPOOLED="postgresql://USER:PASSWORD@HOST/DATABASE?sslmode=require"

# Invite codes for registration. Share them only with the right people.
GUEST_INVITE_CODE="change-me-guest-code"
HOST_INVITE_CODE="change-me-host-code"

# Optional: OpenRouter (https://openrouter.ai/keys), used by "Write it for me".
# Without these the site works, but "Write it for me" says it isn't set up.
OPENROUTER_API_KEY="sk-or-your-key-here"
# Free models only (their id ends in ":free"), so the AI never costs anything.
# Find them at https://openrouter.ai/models (filter on free). Separate several with commas:
# if the first one is busy, OpenRouter tries the next.
OPENROUTER_MODEL="google/gemma-4-31b-it:free,nvidia/nemotron-3.5-lightning:free"

```

Saved in

- .env on my computer
- Vercel for the live site
- .gitignore keeps .env off Github

SUM UP

Server in the middle

Hash passwords

Manage sessions

Use .env